

Excel Functions

Lesson 26 — Core Functions and Error Values

JKSSB Computer Awareness

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From Formulas to Functions

- ▶ A **formula** is an expression that starts with = and calculates a value, for example =A1+A2.
- ▶ A **function** is a ready-made, named operation such as SUM or AVERAGE that performs the calculation for you.
- ▶ A function is used **inside** a formula; its name is followed by arguments in parentheses.
- ▶ Using a function is faster and less error-prone than typing every cell into the formula by hand.

The one idea

A formula is the whole calculation; a function is a built-in tool that performs one part of it.

Function vs Formula

Point	Formula	Function
What it is	Any calculation starting with =	A predefined named operation
Example	=A1+A2	=SUM(A1:A10)
Contains a function?	Not always	Always (by definition)

Trap alert

Every formula must begin with =. Without it, Excel treats the entry as plain text.
(TRAP-30)

Anatomy of a Function

- ▶ A function has a **name** and **arguments** inside parentheses:
`=FUNCTION(argument1, argument2)`.
- ▶ The **name** tells Excel what to do — SUM, MAX.
- ▶ The **arguments** tell it which cells or values to use, such as the range A1:A10.
- ▶ A **range** is written with a colon: A1:A10 means the cells from A1 down to A10.

Worked example

`=SUM(B2:B6)` adds the values in cells B2, B3, B4, B5, and B6.

SUM: Adding Values

- ▶ SUM returns the **total** of the numbers in its arguments.
- ▶ Example: `=SUM(A1:A10)` adds all the numbers in the range A1:A10.
- ▶ Text and blank cells in the range are ignored; only numbers are added.
- ▶ Several ranges can be listed: `=SUM(A1:A5, C1:C5)`.

Remember

SUM means addition — it is the function behind the AutoSum button.

AVERAGE: The Mean

- ▶ AVERAGE returns the **arithmetic mean** of the numbers in a range.
- ▶ Example: `=AVERAGE(A1:A10)` adds the numbers and divides by how many numbers there are.
- ▶ Blank cells and text are **not counted** in the divisor.
- ▶ Trap: it divides by the number of **numeric** cells, not by the total number of cells.

Worked example

If A1:A5 holds 10, 20, 30 and two blanks, `=AVERAGE(A1:A5)` is $(10 + 20 + 30)/3 = 20$, not $60/5$.

COUNT vs COUNTA: The Near-Neighbour

Point	COUNT	COUNTA
Counts	Only cells holding numbers	All non-empty cells
Text cells	Ignored	Counted
Blank cells	Ignored	Ignored
Example	=COUNT(A1:A10)	=COUNTA(A1:A10)

COUNT counts numbers; COUNTA counts anything that is not empty.

COUNT vs COUNTA in Action

- ▶ Suppose A1:A10 holds **6 numbers**, **2 text** labels, and **2 blank** cells.
- ▶ =COUNT(A1:A10) = **6** (numbers only).
- ▶ =COUNTA(A1:A10) = **8** (6 numbers + 2 text labels).
- ▶ Both ignore the blanks; only COUNTA counts the text.

Trap alert

COUNT counts **numbers only**; COUNTA counts **all non-empty** cells. Do not swap this near-neighbour pair.

MAX and MIN

- ▶ MAX returns the **largest** number in a range.
- ▶ MIN returns the **smallest** number in a range.
- ▶ Examples: =MAX(C1:C10) and =MIN(C1:C10).
- ▶ Blank cells and text are ignored, exactly as in SUM and AVERAGE.

Pair them

MAX → biggest; MIN → smallest. Both work on numbers only.

IF: Making a Decision

- ▶ IF tests a condition and returns one of two values.
- ▶ Syntax: `=IF(logical_test, value_if_true, value_if_false)`.
- ▶ Example: `=IF(A1>50,"Pass","Fail")` returns Pass when A1 is more than 50.
- ▶ The first argument must be a **logical test** that is either true or false.

Worked example

`=IF(B2>=40,"Pass","Fail")` shows Pass for a mark of 40 or above and Fail below it.

Nested IF and Blank Input

- ▶ A **nested IF** places an IF inside the value-if-false part, to test more than one condition.
- ▶ Example: `=IF(A1>50,"High",IF(A1>25,"Medium","Low"))`.
- ▶ Excel checks the conditions **in order**; the first true test wins.
- ▶ In a numeric comparison Excel reads a **blank** cell as **0**, so a blank fails the tests and the final value is returned.

PYQ alert

With A1 blank, the formula above returns **Low**: the blank is read as 0. (PYQ-14, JA 2026 Q74)

Other Basic Functions Worth Knowing

Function	What it does
COUNTIF	Counts cells meeting one condition
SUMIF	Adds cells meeting one condition
MEDIAN	Returns the middle value of the numbers
ROUND	Rounds a number to a set number of digits
LEN	Counts the characters in a text value
TODAY	Returns today's date

At JKSSB depth, know what each one returns and that all are written inside a formula.

AutoSum

- ▶ **AutoSum** is a button that inserts a SUM formula for the cells beside the active cell.
- ▶ By default it writes =SUM(...) over the range Excel guesses you want.
- ▶ Keyboard shortcut: **Alt + =** inserts AutoSum.
- ▶ Its drop-down also offers Average, Count Numbers, Max, and Min.

Shortcut alert

AutoSum is **Alt + =**, not a Ctrl combination or a function key. This is a common wrong option.

Excel Error Values (1)

Error	Meaning
#DIV/0!	A number is divided by zero or by a blank cell
#VALUE!	The wrong type of argument — text where a number is needed
#NAME?	Excel does not recognise a name — often a misspelt function name

Read the error as a clue to what Excel could not do.

Excel Error Values (2)

Error	Meaning
#REF!	A cell reference is not valid — the referenced cell was deleted
#N/A	The value is not available, as when a lookup finds no match
#NUM!	A number in the formula is not valid for the operation

is not an error value — it only means the column is too narrow to show the number.

Trap Alert: What to Keep Straight

- ▶ A formula must start with =; without it the entry is text. (TRAP-30)
- ▶ COUNT counts **numbers**; COUNTA counts all **non-empty** cells.
- ▶ AVERAGE divides by the number of numeric cells, not by all the cells.
- ▶ A **function** is a named built-in operation used **inside** a formula.
- ▶ #DIV/0! is division by zero; #VALUE! is a wrong data type; #NAME? is an unrecognised name.

Office Desktop Example

- ▶ A marks sheet lists student names in A2:A31 and marks in B2:B31.
- ▶ `=COUNTA(A2:A31)` counts how many students were entered.
- ▶ `=COUNT(B2:B31)` counts how many have a numeric mark.
- ▶ `=AVERAGE(B2:B31)` gives the class average; `=MAX(B2:B31)` gives the top mark.

Read the two counts together

If COUNTA gives 30 but COUNT gives 28, two students have a non-numeric entry in the marks column — a data-entry problem to fix.

Lesson 26: Recall

- ▶ A formula starts with =; a function such as SUM or IF is a named operation used inside a formula.
- ▶ SUM adds, AVERAGE means, MAX/MIN give the largest/smallest number.
- ▶ COUNT counts numbers; COUNTA counts all non-empty cells — the key near-neighbour pair.
- ▶ IF(logical_test, value_if_true, value_if_false); nested IF checks conditions in order, and a blank is read as 0 in a numeric test (PYQ-14).
- ▶ AutoSum inserts =SUM(...); the shortcut is Alt + =.
- ▶ Error values: #DIV/0!, #VALUE!, #NAME?, #REF!, #N/A.